Course 1: LoRa & LoRaWAN

Technical Deep Dive

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Course Overview

What You'll Learn

- LoRa Physical Layer
  - Chirp Spread Spectrum modulation
  - Link budget calculations
  - Time-on-Air analysis
- LoRaWAN Protocol
  - MAC layer architecture
  - Device classes (A, B, C)
  - Join procedures (OTAA, ABP)
- Hands-On Labs
  - Lab 1.1: Link budget calculator
  - Lab 1.2: Time-on-Air analysis
  - Lab 1.3: OTAA join implementation
  - Lab 1.4: Class C deployment

What is LoRa?

Long Range Radio Technology

Definition

LoRa is a proprietary chirp spread spectrum (CSS) modulation technique developed by Semtech.

Key Features:

- Long range (2-15 km)
- Low power (< 100 µA sleep)
- Low data rate (0.3-50 kbps)
- Robust to interference

Applications:

- Smart cities
- Agriculture sensors
- Asset tracking
- Utility metering

Proprietary Note

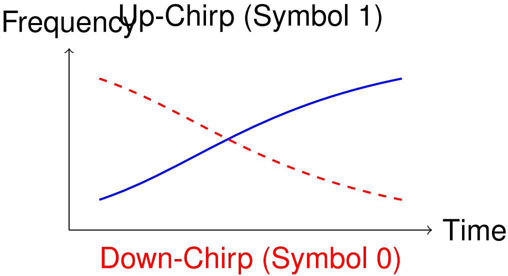
LoRa PHY is proprietary (Semtech), but LoRaWAN MAC layer is open standard (LoRa Alliance).

Chirp Spread Spectrum

How LoRa Modulates Data

Definition (Chirp)

A chirp is a signal whose frequency increases (up-chirp) or decreases (down-chirp) linearly with time.



Encoding Data

Different data symbols are represented by chirps starting at different

LoRa Parameters

Configuration Options

Parameter	Description
Spreading Factor (SF)	7-12. Higher SF = longer range, lower data rate
Bandwidth (BW)	125, 250, 500 kHz. Wider BW = higher data rate
Coding Rate (CR)	4/5, 4/6, 4/7, 4/8. More redundancy = more robust
TX Power	2-20 dBm (EU), 2-30 dBm (US). Higher = longer range
Frequency	ISM bands: 868 MHz (EU), 915 MHz (US), 433 MHz (Asia)

Theorem (Spreading Factor Trade-off)

Increasing SF by 1 doubles the range but halves the data rate and doubles Time-on-Air.

Link Budget Calculation

Estimating Maximum Range

Link Budget Formula

$$RX_{power} = TX_{power} + TX_{gain} - PathLoss + RX_{gain} - Margin$$

Free Space Path Loss (FSPL):

$$FSPL(dB) = 20\log_{10}(d_{km}) + 20\log_{10}(f_{MHz}) + 32.45$$

Example Calculation

- TX Power: +14 dBm (EU868 limit)
- TX/RX Antenna Gain: 2 dBi each
- Frequency: 868 MHz
- Distance: 10 km
- LoRa Sensitivity (SF12): -137 dBm

$$FSPL = 20\log(10) + 20\log(868) + 32.45 = 131.4 \text{ dB}$$

$$RX = 14 + 2 - 131.4 + 2 = -113.4 \text{ dBm}$$

Time-on-Air (ToA)

Airtime = Duty Cycle Compliance

Why ToA Matters

EU868 regulation: Maximum 1% duty cycle per sub-band per day (36 seconds/hour).

ToA Formula (simplified):

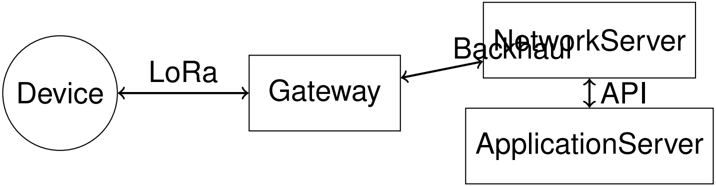
$$ToA \approx \frac{2^{SF} \cdot PayloadBytes}{BW \cdot CR}$$

SF	ToA (10 bytes)	Max msgs/hour @ 1%
SF7	41 ms	878
SF8	72 ms	500
SF9	144 ms	250
SF10	247 ms	145
SF11	494 ms	72
SF12	991 ms	36

Duty Cycle Violation

LoRaWAN Architecture

End-to-End Network



End Device: Battery-powered sensor transmitting data

Gateway: Receives LoRa packets, forwards to network server via IP

Network Server: Handles MAC layer, de-duplication, ADR

Application Server: Your application logic, data storage

Device Classes

Class A, B, and C

Class A

- Lowest power
- TX + 2 RX windows
- Downlink only after uplink
- Latency: minutes

Class B

- Scheduled RX slots
- Beacon synchronization
- Lower latency (seconds)
- Higher power

Class C

- Continuous RX
- Lowest latency (instant)
- Highest power
- Mains-powered

Class A Diagram

Class B Diagram

Class C Diagram

OTAA vs ABP

Device Activation Methods

OTAA (Recommended)

- Over-The-Air Activation
- Join procedure
- Dynamic session keys
- Secure
- Recovers from network reset

ABP (Legacy)

- Activation By Personalization
- Pre-provisioned keys
- Static session keys
- Less secure
- Frame counter issues

```
// OTAA credentials
uint8_t DevEUI[8] = {0x...};
uint8_t AppKey[16] = {0x...};

// Join network
smtc_modem_join_network(0);

// Wait for JOIN_ACCEPT
```

```
// ABP credentials
uint32_t DevAddr = 0x...;
uint8_t NwkSKey[16] = {0x...};
uint8_t AppSKey[16] = {0x...};

// No join needed
smtc_modem_set_app_skey(...);
smtc_modem_set_nwk_skey(...);
```

Recommendation

Always use OTAA for production deployments. ABP is for testing only.

Adaptive Data Rate (ADR)

Automatic Link Optimization

What is ADR?

Network server instructs device to adjust spreading factor and TX power based on link quality.

Benefits:

- Maximizes network capacity
- Minimizes power consumption
- Reduces collisions

ADR Algorithm:

1. Device reports uplink with current DR and power
2. Network server monitors RSSI/SNR over 20 uplinks
3. If link margin > 15 dB, increase DR or reduce power
4. If link margin < 5 dB, decrease DR or increase power

When to Disable ADR

Disable ADR for mobile devices (vehicles, asset trackers) as link

Lab 1.1: Link Budget Calculator

Python Implementation

Objective: Calculate maximum range for different LoRa configurations.

Deliverables:

- Python script implementing FSPL formula
- Test scenarios: SF7, SF10, SF12 at various distances
- Comparison table: Urban vs Rural vs Suburban

Example Output:

```
Distance: 10 km, Frequency: 868 MHz
  SF7  (Sens: -123 dBm): Margin = 7.6 dB  [PASS]
  SF10 (Sens: -132 dBm): Margin = 16.6 dB [PASS]
  SF12 (Sens: -137 dBm): Margin = 21.6 dB [PASS]
```

Learning Outcome: Understand relationship between distance, SF, and link budget.

Lab 1.2: Time-on-Air Analysis

Compliance Calculator

Objective: Ensure duty cycle compliance for EU868.

Tasks:

1. Calculate ToA for different payload sizes
2. Determine maximum messages per hour (1% duty cycle)
3. Analyze impact of confirmed vs unconfirmed uplinks

```
def calculate_toa(sf, bw, payload_bytes, cr=1):
    """Calculate Time-on-Air in milliseconds"""
    symbol_time = (2**sf) / bw
    preamble_time = (8 + 4.25) * symbol_time

    payload_symbols = 8 + max(
        ceil((8*payload_bytes - 4*sf + 28) / (4*sf)) * (cr + 4),
        0
    )

    payload_time = payload_symbols * symbol_time
    return (preamble_time + payload_time) * 1000  # ms
```

Learning Outcome: Design compliant uplink schedules.

Lab 1.3: OTAA Join Implementation

First LoRaWAN Application

Objective: Join network and send periodic uplinks.
Hardware: Xiao-nRF54L15 + LR1120 shield

Code Skeleton:

```c
int main(void) {
    // Initialize modem
    smtc_modem_init(event_handler);
    smtc_modem_set_region(STACK_ID, SMTC_MODEM_REGION_EU_868);

    // Start join
    smtc_modem_join_network(STACK_ID);

    while (1) {
        smtc_modem_run_engine();

        if (joined && time_for_uplink()) {
            uint8_t payload[] = {0x01, 0x02, 0x03};
            smtc_modem_request_uplink(STACK_ID, 2, false,
                                      payload, 3);
        }

        k_msleep(100);
    }
}
```

Learning Outcome: Understand LoRaWAN join flow and uplink

Lab 1.4: Class C Implementation

Continuous Receive Mode

Objective: Switch to Class C for immediate downlinks.

Modifications:

1. After join, switch to Class C:
   - `smtc`$_{m}odem_{s}et_{c}lass(STACK_{I}D, SMTC_{M}ODEM_{C}LASS_{C})$;
2. Handle downlinks in `SMTC_MODEM_EVENT_DOWNDATA`
3. Measure power consumption difference

Expected Results:

Metric	Class A	Class C
Avg Current	120 µA	15 mA
Downlink Latency	60 seconds	< 1 second
Battery Life (2000 mAh)	694 days	5.6 days

Learning Outcome: Understand Class C trade-offs: latency vs power.

Course 1 Summary

Key Takeaways

You've Learned:

- LoRa PHY: CSS modulation, SF/BW/CR parameters
- Link Budget: Calculate max range using FSPL
- Time-on-Air: Ensure regulatory compliance
- LoRaWAN: MAC layer, device classes, OTAA
- Practical Skills: Build, flash, and test LoRaWAN devices

Next Steps:

- Course 2: LoRa Basics Modem Architecture
- Course 3: USP RAC for Multiprotocol
- Explore advanced features: FUOTA, Relay, Geolocation

Assessment

Complete all 4 labs and pass the final quiz (80%+) to earn Course 1 certificate.

Thank You!

Questions?